

MILESTONE 2

End-to-End Data Analysis & Visualization Project

Group Project Guide & Grading Rubric

Group size: 3–4 students

Format: Applied, self-directed project

Builds on: (1) Data Wrangling & Cleaning with Pandas/NumPy, and (2) Exploratory Data Analysis, Statistical Reasoning & Visualization

1. Overview

In Week 5-7, you learned the individual skills needed to clean messy data and explore it statistically. In Milestone 2, your group will put those skills together into a single, self-directed, end-to-end analysis — from finding a real dataset, to cleaning it, to exploring it, to communicating what you found to someone who has never seen the data.

There is no "correct" dataset or "correct" question. You are being assessed on process, judgment, and communication — not on getting a dramatic result.

What "end-to-end" means here

- Find a real, messy dataset (or combination of datasets) relevant to a question your group cares about.
- Clean it, documenting every decision.
- Explore it statistically and visually.
- Interpret what you found, honestly and with appropriate caveats.
- Communicate it to two audiences: a technical one (your notebook/code) and a non-technical one (your presentation).

2. Learning Objectives

By the end of this milestone, each group member should be able to:

- Independently source and evaluate a real-world dataset for suitability and quality
- Diagnose and resolve data quality issues (missing values, duplicates, inconsistent types, encoding issues, outliers) and justify the chosen strategy over the alternatives
- Merge, join, or reshape multiple data sources into one working analytical dataset
- Compute and correctly interpret descriptive statistics and basic inferential concepts (sampling, confidence, significance)
- Conduct a structured EDA rather than ad hoc plotting
- Select and build the right visualization for a given analytical question, while avoiding common statistical/visual pitfalls
- Communicate findings clearly to a non-technical reader, including the limitations of the analysis
- Collaborate effectively in a small technical team with shared version control

3. Project Structure & Deliverables

Part	Deliverable	Format
A	Project Proposal	1-page doc or slide
B	Cleaned Dataset + Cleaning Log	Jupyter notebook (01_cleaning.ipynb) + cleaned .csv/.parquet
C	EDA & Statistical Analysis	Jupyter notebook (02_eda.ipynb)
D	Visualization Set	4–6 finished charts embedded in report (.png/.svg + notebook)
E	Presentation	8–10 minute group presentation + slides
F	Contribution Log	Shared doc listing who did what

All notebooks and the cleaned dataset should live in a single shared repository or folder submitted at the end.

4. Step-by-Step Guide

Step 1 — Form Groups & Choose a Question

- Groups of 3–4. Assign a rough division of labor early, but everyone should touch cleaning, EDA, and communication at least once — this is graded individually (see Section 6).
- Pick a real-world question you can actually answer with data you can find. Good questions are specific and comparative/relational, not vague.
 - Weak: "Analyze video game sales."
 - Strong: "Has the average review score for indie games changed relative to AAA games over the last decade, and does price explain any of the difference?"

Step 2 — Find Your Data (you are sourcing this yourselves)

You must find and justify your own dataset(s). This is part of the assessment — dataset selection is a real-world data science skill.

Where to look

- Government/public open data portals (e.g., national statistics offices, city/state open data portals, World Bank, WHO, data.gov-style portals)
- Kaggle, UCI Machine Learning Repository, Google Dataset Search
- Public APIs (weather, sports, finance, transit, social platforms with public endpoints)
- Organizational or NGO published datasets/reports with downloadable data

Requirements for your dataset(s)

- At least one dataset must be genuinely messy (missing values, inconsistent formatting, mixed types, duplicates, or similar) — not a pre-cleaned "textbook" dataset
- Prefer combining two or more sources that need to be merged/joined, but a single sufficiently rich and messy dataset is acceptable if merging genuinely isn't possible for your question
- Enough rows/columns to support meaningful EDA (as a rule of thumb, avoid trivially small datasets, e.g., under a few hundred rows, unless the domain justifies it)
- Legally and ethically usable (public, licensed for reuse, no scraping of personal/private data)

Step 2 deliverable: a short proposal stating your question, your data source(s) with links, why the data is suitable (and messy), and any known risks (e.g., a source might not have enough data, might require an API key, etc.)

Step 3 — Clean & Wrangle

Using your Week 5-6 skills:

- Load and inspect the data; profile it (shape, dtypes, missingness, duplicates)
- Diagnose specific data quality issues present in your dataset — don't apply a generic checklist blindly
- Choose and justify cleaning strategies (deletion vs. imputation, and which imputation method, if any) — explain trade-offs, not just what you did
- Fix data types, remove/handle duplicates, handle outliers deliberately (don't silently drop them without justification)
- Merge/join/concatenate and reshape (melt/pivot) as needed to build one analysis-ready table
- Handle formatting/encoding issues (e.g., inconsistent categories, string casing, encoding errors) if present
- Document every decision in markdown cells: what the issue was, what you did about it, and why — this log is graded

Step 4 — Exploratory Data Analysis

Treat EDA as a structured process, not random plotting:

- Univariate: distributions of key variables (shape, central tendency, spread)
- Bivariate/multivariate: relationships between variables relevant to your question (correlation, group comparisons)
- Outlier and anomaly review
- Segment comparisons relevant to your research question

Step 5 — Apply Statistical Reasoning

- Report the descriptive statistics that matter for your question (not everything you can compute)
- Where relevant, apply basic inferential ideas: sampling considerations, confidence intervals, or a simple significance test — and explain, in plain language, what the result does and doesn't mean
- Explicitly check your own analysis for common pitfalls: misleading axes, confusing correlation with causation, cherry-picking supportive subsets, small-sample overconfidence

Step 6 — Build Visualizations

- Choose chart types based on the question being asked, not habit (e.g., distribution → histogram/box/violin; relationship → scatter; comparison → bar/grouped bar; trend → line)

- Each chart needs: a clear title, labeled axes with units, an honest scale (no truncated/misleading axes), and a one-sentence caption stating the takeaway
- Favor a small number of clear, well-labeled charts (aim for 4–6) over many cluttered ones

Step 7 — Present & Peer Review

- 8–10 minutes, all group members or representatives speak
- Walk through the question, key findings, and one limitation you'd want to fix with more time
- Be ready for questions probing your cleaning decisions and whether your conclusions are actually supported by your evidence

5. Suggested Timeline

Week	Milestone
6	Groups formed, question chosen, dataset(s) sourced, proposal submitted
6	Data loaded, profiled, cleaning underway
7	Cleaning finalized; EDA begins
7	EDA and statistical analysis complete; visualizations drafted
8	Report written, visualizations finalized, presentation built
8	Presentations, peer review, final submission

6. Grading Rubric (GROUP) — 100 points total

Each criterion is scored at one of four performance levels. Partial credit within a level is allowed.

A. Data Sourcing & Proposal — 10 pts

Level	Points	Description
Exemplary	9–10	Data is genuinely messy, well-justified, appropriately scoped, and clearly tied to a specific, answerable question. Sourcing risks anticipated.
Proficient	7–8	Data is suitable and justified, question is reasonably specific.
Developing	4–6	Data or question is vague, too clean, or too broad to support real analysis.
Beginning	0–3	No clear question, or data is unsuitable/unusable.

B. Data Cleaning & Wrangling — 20 pts

Level	Points	Description
Exemplary	18–20	Systematically diagnoses real issues in the data; chosen strategies (imputation/deletion/etc.) are justified with trade-offs discussed; merges/reshapes multiple sources correctly; every decision documented for reproducibility.
Proficient	14–17	Diagnoses and resolves most real issues with reasonable justification; documentation present but less thorough.
Developing	8–13	Applies generic cleaning steps without clear diagnosis or justification; some issues missed; documentation thin.
Beginning	0–7	Little evidence of systematic cleaning; undocumented or largely absent.

C. EDA & Statistical Reasoning — 20 pts

Level	Points	Description
Exemplary	18–20	Structured EDA (univariate → bivariate → segments → outliers); correctly computes and interprets descriptive stats; applies an inferential concept correctly and explains it in plain language; explicitly identifies pitfalls.
Proficient	14–17	Solid structured EDA and mostly correct statistical interpretation; inferential reasoning present but shallow.
Developing	8–13	EDA is largely descriptive/ad hoc; statistical claims are present but not well interpreted or occasionally incorrect.
Beginning	0–7	Minimal or unstructured EDA; statistics misapplied or absent.

D. Visualization Quality — 20 pts

Level	Points	Description
Exemplary	18–20	Chart types clearly matched to the question; all charts correctly labeled, honestly scaled, and captioned with a clear takeaway; 4–6 charts, no clutter or redundancy.
Proficient	14–17	Good chart choices, minor labeling/scaling issues, mostly clear takeaways.
Developing	8–13	Charts present but mismatched to question, missing labels/units, or misleading in places (e.g., truncated axes).
Beginning	0–7	Few or no meaningful charts; unclear or misleading.

E. Narrative Report — 20 pts

Level	Points	Description
Exemplary	18–20	Clear, accurate story for a non-technical reader; findings tie directly to visuals; limitations and alternative explanations honestly discussed; direct, appropriately hedged conclusion.
Proficient	14–17	Clear and mostly accurate; limitations addressed but could be more specific.
Developing	8–13	Report is technical/jargon-heavy or overstates findings; limitations thin or missing.
Beginning	0–7	Missing, incoherent, or unsupported by the actual analysis.

F. Code Quality & Reproducibility — 10 pts

Level	Points	Description
Exemplary	9–10	Notebooks run top-to-bottom without errors; clean, organized, commented code; clear separation of cleaning/EDA steps; a stranger could reproduce the analysis.
Proficient	7–8	Runs with minor issues; reasonably organized and commented.
Developing	4–6	Runs with effort; disorganized or poorly commented.
Beginning	0–3	Does not run, or unusable/undocumented.

Total: ____ / 100

7. Grading Rubric (INDIVIDUAL) — 50 points total

Each criterion is scored at one of four performance levels. Partial credit within a level is allowed.

A. Presentation Proper — 25 pts

Level	Points	Description
Exemplary	23–25	Clear, well-paced, all members speak, answers questions about methodology confidently and honestly.
Proficient	20–22	Clear and organized; most members participate.
Developing	15–19	Disorganized or uneven participation.
Beginning	0–14	Unprepared or largely absent presentation.

B. Teamwork & Individual Contribution — 25 pts

Level	Points	Description
Exemplary	23–25	Clear, substantive, roughly even contribution across cleaning, analysis, and communication; log is specific.
Proficient	20–22	Solid contribution, slightly uneven.
Developing	15–19	Contribution concentrated in one area or one person; log is vague.
Beginning	0–14	Little evidence of individual contribution.

Total: ____ / 50

8. Common Pitfalls to Avoid

- Choosing a dataset that's already clean — you'll have nothing to demonstrate in Section B
- Dropping missing values or outliers without explaining why, or without considering alternatives
- Plotting everything you can instead of what answers the question
- Truncated/non-zero-baseline axes that exaggerate differences
- Stating correlation as causation
- A report full of code and jargon instead of a plain-language narrative
- Skipping the limitations section — every real analysis has them, and naming yours is a sign of rigor, not weakness
- One person doing all the cleaning/coding while others only write the report (or vice versa)

9. Submission Checklist

- ☐ Proposal (Part A)
- ☐ Cleaning notebook + cleaned dataset file (Part B)
- ☐ EDA/statistics notebook (Part C)
- ☐ 4–6 finished, labeled, captioned visualizations (Part D)
- ☐ Presentation slides (Part E)
- ☐ Contribution log signed off by all group members (Part F)
- ☐ All notebooks run top-to-bottom without errors
- ☐ Data source(s) and licensing/usage terms cited